

■ 1.4.11 Tagging Objects with the Names of Symbols

There are many ways to use symbols in *Mathematica*. So far we have concentrated on using symbols to store values, and to represent mathematical variables. This section describes another way to use symbols in *Mathematica*.

The basic idea is to use symbols as “tags” for different types of objects.

Working with physical units gives one simple example. When you specify the length of an object, you want to give not only a number, but also the units in which the length is measured. In standard notation, you might write a length as 12 meters.

You can imitate this notation almost directly in *Mathematica*. You simply use a symbol `meters` to indicate the units of our measurement.

The symbol `meters` here acts as a “tag”, which indicates the units used.

```
In[1]:= 12 meters
Out[1]= 12 meters
```

You can add lengths like this.

```
In[2]:= % + 5.3 meters
Out[2]= 17.3 meters
```

This gives a speed.

```
In[3]:= % / (25 seconds)
Out[3]=  $\frac{0.692 \text{ meters}}{\text{seconds}}$ 
```

This converts to a speed in feet per second.

```
In[4]:= % /. meters -> 3.28084 feet
Out[4]=  $\frac{2.27034 \text{ feet}}{\text{seconds}}$ 
```